Senthil Nagappan

Forward-Deployed AI Product Engineer · Human-Centered AI, Accessibility & Human Factors

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PROFESSIONAL SUMMARY

Forward-deployed AI product engineer with 18+ years working inside complex, regulated workflows — healthcare, federal, and consumer health — alongside users, engineering, product, risk, and client teams. Combines AI governance, accessibility, design systems, Human Factors, and front-end prototyping to make AI usable, accountable, and measurable. 7+ years in compliance, risk, and advisory delivery; directed a $130M federal behavioral-health program portfolio and scaled a cross-functional team to 15+.

FORWARD-DEPLOYED OPERATING MODEL

Embed in real workflows · **Frame** user, technical, and risk constraints · **Prototype** with engineering · **Integrate** inside the release cadence · **Validate** adoption, override, correction, and audit evidence.

CORE CAPABILITIES

Forward-deployed AI delivery · Field immersion & workflow discovery · Cross-functional problem framing · Engineering partnership · Prototyping (React/TypeScript) · Deployment instrumentation · Client & executive delivery · AI governance & model risk · NIST AI RMF · EU AI Act · ISO/IEC 42001 · Human–AI interaction · Explainability & trust UX · Human-in-the-loop oversight · WCAG 2.2 · Section 508 · Design systems · IEC 62366 · MIL-STD-1472H

EXPERIENCE

AI Experience Designer & HSI Lead

GE HealthCare (Brain Pulse Inc) · Mar 2025 – Present · Remote

- Work forward-deployed with product and engineering teams, translating field context, clinical risk, and governance requirements into release-ready product behavior.
- Design agentic clinical workflows where the model recommends and the clinician decides, with oversight tiers, override paths, source grounding, and known-limits disclosure.
- Prototype in React and TypeScript; design accessible design-system components adopted across 10+ teams and map behavior to NIST AI RMF and IEC 62366 evidence.

Sr. UX & Accessibility Lead

Social Security Administration (Leidos) · Oct 2022 – Feb 2025 · Remote

- Worked forward-deployed inside federal program delivery, connecting caseworker needs, engineering constraints, accessibility policy, and 40+ monthly releases.
- Led and governed an enterprise design system across 15+ modules (platform scope: 50M+ citizens). Personally traced 65% of interaction defects to 8 reused components; fixing them at the source contributed to a ~30% manual-audit reduction and an estimated $1.5M+ in annual savings (program outcomes).
- As product owner, led the introduction of AI-assisted WCAG 2.2 AA / Section 508 validation as a CI/CD release gate — defined the human review workflow and gate criteria with engineering, who integrated it. Scaled and mentored a team to 15+.

Human Factors Engineer · Sr. UX & Inclusive Design Lead

Best Buy Health · Sep 2021 – Sep 2022 · Remote

- Embedded field research and use-related risk analysis into product, engineering, and validation decisions for connected medical-device experiences (program scope: 1M+ users); contributed to a 30–40% drop-off reduction (team outcome).
- Ran summative usability validation supporting 510(k) human-factors evidence under FDA HFE, IEC 62366, and ISO 14971; contributed accessible BRIX design-system patterns.

Sr. UX & Program Lead

HHS / SAMHSA (Eagle Technologies) · Mar 2014 – May 2021 · Remote

- Directed a $130M portfolio of federal behavioral-health programs, leading a client-delivery model spanning discovery, roadmap, stakeholder alignment, team growth, and release governance for national data services supporting 2M+ users.

Earlier Roles

AAMC · CareFirst BCBS · JPMorgan Chase · Morgan Stanley · 2007 – 2014

- Enterprise UX, accessibility, front-end delivery, and financial-services work that established the regulated-delivery and model-risk discipline now applied to AI.

SELF-DIRECTED AI PROTOTYPES (NOT CLIENT WORK)

- **Visual-assistance app** — prototype I built and host as a public web app (React/TypeScript, serverless proxy to an LLM vision API): scene description, OCR, product identification. No documented users or clinical use.
- **Lumen** — RAG enterprise knowledge assistant with source citations and evaluation; in build.
- **TrustLens** — concept for AI review with approve / reject / escalate, audit trail, and evaluation suite.

EDUCATION & CREDENTIALS

B.E., Electronics & Communication Engineering (HCI / human-factors focus), Madurai Kamaraj University, 1999.
AIGP — AI Governance Professional (IAPP), in progress · IAAP CPACC + WAS (CPWA) · CUA · CSM · Salesforce Certified UX Designer, in progress · DAU HSI, in progress

STANDARDS (APPLIED OR WORKING KNOWLEDGE) & COMMUNITY

Applied: WCAG 2.2 AA · Section 508 · FDA HFE / IEC 62366 · ISO 14971 · MIL-STD-1472H. Working knowledge: MBSE/SysML · DoDI 5000.95 · NIST AI RMF · EU AI Act · ISO/IEC 42001 · SR 11-7
Vice President, Texas Astronomical Society; Cubmaster & Assistant Scoutmaster, Scouting America; Lead Coach, FIRST LEGO League.